Differentiable Welfare Theorems
Existence of a Competitive Equilibrium: Preliminaries

Econ 2100 Fall 2018

Lecture 19, November 5

Outline

1. Welfare Theorems in the differentiable case.
2. Aggregate excess demand and existence.

Welfare Theorems in the Differentiable Case

Question

What is the relationship between the first order conditions that correspond to a competitive equilibrium and those that give Pareto optimality?

Assumptions needed for differentiability

- Consumers
- Producers

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  - Normalize things so that $u_i(0) = 0$.
  - Assume each $u_i(x)$ is twice continuously differentiable, with $\nabla u_i(x) \gg 0$ for any x, and also assume that $u_i(x)$ is quasi-concave.
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- Producers
  - Production sets are $Y_j = \left\{ y \in \mathbb{R}^L : F_j(y) \leq 0 \right\}$, where $F_j(y) = 0$ defines the transformation frontier.
  - Assume each $F_j(y)$ is convex, twice continuously differentiable, with $\nabla F_j(y) \gg 0$ for any y, and also assume that $F_j(0) \leq 0$.

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Welfare Theorems in the Differentiable Case

- Given these assumptions, Pareto efficiency solves the planner's problem.

Remark

An allocation is Pareto optimal if and only if it is a solution to the following:

$$\max_{(x,y)\in\mathbb{R}^{LI}\times\mathbb{R}^{LJ}} u_1\left(x_{11}, x_{21}, ..., x_{L1}\right)$$

subject to

$$\begin{aligned} u_i\left(x_{1i}, x_{2i}, ..., x_{Li}\right) &\geq \bar{v}_i && i = 2, 3, ..., I \\ \sum_i x_{li} &\leq \omega_l + \sum_j y_{lj} && l = 1, 2, ..., L \\ F_j\left(y_{1j}, ..., y_{Lj}\right) &\leq 0 && j = 1, 2, ..., J \end{aligned}$$

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Welfare and Equilibirum In the Differentiable Case

$(\hat{x}, \hat{y})$ is Pareto optimal if and only if it solves the following

$$\max_{(x,y)\in\mathbb{R}^{L(I+J)}} u_1(x_1) \quad s.t. \quad \begin{array}{ll} u_i(x_i) \geq \bar{v}_i & i = 2,3,...,I \\ F_j(y_j) \leq 0 & j = 1,2,...,J \\ \sum_i x_{li} \leq \omega_l + \sum_j y_{lj} & l = 1,2,...,L \end{array}$$

- We can also write the maximization problems that must be solved by a competitive equilibrium.

A competitive equilibrium with transfers x^*, y^*, p^* solves the following $I + J$ optimization problems (with $w_i = p^* x_i^*$ for all i):

$$\max_{x_i \geq 0} u_i(x_i) \quad s.t. \quad p^* \cdot x_i \leq w_i \quad i = 2,3,...,I$$

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The first order conditions for a Pareto optimum are:

$$x_{li} \quad : \quad \delta_i \frac{\partial u_i(x_i^*)}{\partial x_{li}} - \mu_l \begin{cases} \leq 0 \\ = 0 \quad \text{if } x_{li} > 0 \end{cases} \qquad \text{for all } i \text{ and } l$$

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where, to make the notation more compact, $\delta_1 = 1$.

- With convexity assumptions, these conditions are also sufficient.

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Interpret the multipliers as follows:

- μ_l is the change in *consumer 1* utility from a marginal increase in the aggregate endowment of good l (this the "shadow price" of good l).
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- Thus $\gamma_j \frac{\partial F_j}{\partial y_{lj}}$ is the marginal cost of increasing y_{lj}; it must be equal to the marginal benefit of good l (measured in consumer 1 utility).

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"Marginal" Conditions for Pareto Optima

- At an interior solution ($x_{li}^* > 0$) we can rewrite these first order conditions as the "usual" equalities between marginal rates of substitution and transformation.

MRS between any two goods must be equal across any two consumers:

$$\frac{\frac{\partial u_i}{\partial x_{li}}}{\frac{\partial u_i}{\partial x_{l'i}}} = \frac{\frac{\partial u_{i'}}{\partial x_{li'}}}{\frac{\partial u_{i'}}{\partial x_{l'i'}}} \qquad \text{for all } i, i' \text{ and } l, l'$$

MRT between any two goods must be equal across any two firms:

$$\frac{\frac{\partial F_j}{\partial y_{lj}}}{\frac{\partial F_j}{\partial y_{l'j}}} = \frac{\frac{\partial F_{j'}}{\partial y_{l'j}}}{\frac{\partial F_{j'}}{\partial y_{l'j'}}} \qquad \text{for all } j, j' \text{ and } l, l'$$

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- There are three types of efficiency embedded in Pareto optimality (what?).

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Equilibrium in the Differentiable Case

A price equilibrium with transfers solves

$$\max_{x_i \geq 0} u_i(x_i) \text{ s.t. } p^* \cdot x_i \leq w_i \; i = 2, 3, ..., I \quad \text{and} \quad \max_{y_j} p^* \cdot y_j \text{ s.t. } F_j(y_j) \leq 0 \; j = 1, ..., J$$

- Let α_i be the Lagrange multiplier associated with i's budget constraint.
- Let β_j be the Lagrange multiplier associated with j's technology constraint.
- The first order necessary and sufficient conditions for these problems yield:

$$x_{li} : \frac{\partial u_i}{\partial x_{li}} - \alpha_i p_l^* \begin{cases} \leq 0 \\ = 0 \quad \text{if } x_{li} > 0 \end{cases} \quad \forall i, l \qquad y_{lj} : p_l^* - \beta_j \frac{\partial F_j}{\partial y_{lj}} = 0 \quad \forall j, l$$

- When the non-negativity constraints bind, the consumers' conditions imply

$$\frac{p_l^*}{p_{l'}^*} = \frac{\frac{\partial u_i}{\partial x_{li}}}{\frac{\partial u_i}{\partial x_{l'i}}} = \frac{\frac{\partial u_{i'}}{\partial x_{li'}}}{\frac{\partial u_{i'}}{\partial x_{l'i'}}} \quad \text{for all } i, i' \text{ and } l, l'$$

- The firms' conditions imply

$$\frac{p_l^*}{p_{l'}^*} = \frac{\frac{\partial F_j}{\partial y_{lj}}}{\frac{\partial F_j}{\partial y_{l'j}}} = \frac{\frac{\partial F_{j'}}{\partial y_{l'j}}}{\frac{\partial F_{j'}}{\partial y_{l'j'}}} \quad \text{for all } j, j' \text{ and } l, l'$$

- These should look familiar.

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An allocation is Pareto optimal if and only if it is a solution to:

$$\max_{(x,y)\in\mathbb{R}^{L(I+J)}} u_1(x_1) \quad s.t. \quad \begin{array}{ll} u_i(x_i) \geq \bar{u}_i & i = 2,3,...,I \\ F_j(y_j) \leq 0 & j = 1,2,...,J \\ \sum_{i=1}^{I} x_{li} \leq \omega_l + \sum_{j=1}^{J} y_{lj} & l = 1,2,...,L \end{array}$$

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Then, any solution to the first optimization problem solves the others.

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Summary

In the case of a differentiable economy, we can use the first order conditions to prove the following.

- Every Pareto optimal allocation is a price equilibrium with transfers for some appropriately chosen price vector p and welfare transfers w (Second Welfare Theorem);
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- As an exercise, write these conclusions like formal 'theorems' rather than 'observations'.
- Note: the price of each good corresponds to its marginal value in the Pareto optimality problem.

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The Other Planner's Problem

An equivalent maximization problem (also called the "Planner's problem")

$$\max_{(x,y)\in\mathbb{R}^{L(I+J)}} \sum_{i=1}^{I} \lambda_i u_i(x_i) \qquad s.t. \qquad \begin{array}{ll} F_j(y_j) \leq 0 & j = 1,2,...,J \\ \sum_{i=1}^{I} x_{li} \leq \omega_l + \sum_{j=1}^{J} y_{lj} & l = 1,2,...,L \end{array}$$

- The first order conditions for this problem are:

$$x_{li}: \lambda_i \frac{\partial u_i}{\partial x_{li}} - \psi_l \begin{cases} \leq 0 \\ = 0 \quad \text{if } x_{li} > 0 \end{cases} \quad \forall i, l$$

- Connect to the other planner's problem:

$$\mu_l = \frac{\psi_l}{\lambda_1}, \quad \delta_i = \frac{\lambda_i}{\lambda_1}, \text{ and } \quad \gamma_j = \frac{\eta_j}{\lambda_1}$$

- Conect to equilibrium:

$$p_l^* = \psi_l, \quad \alpha_i = \frac{1}{\lambda_i}, \text{ and } \quad \beta_j = \eta_j$$

This yields a new interpretation of equilibrium prices at an interior optimum

$\sum_{l=1}^{L} \lambda_i \frac{\partial u_i}{\partial x_{li}} = \sum_{l=1}^{L} \psi_l$ and thus $\frac{\lambda_i \frac{\partial u_i}{\partial x_{li}}}{\sum_{l=1}^{L} \lambda_i \frac{\partial u_i}{\partial x_{li}}} = \frac{\psi_l}{\sum_{l=1}^{L} \psi_l}$ for all l (r.h.s. same for all i).

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Existence of A Competitive Equilibrium

The Existence Problem

- We have theorems about the welfare properties of a competitive equilibrium, but no result stating that such a thing exist.
- Existence is always a crucial question (any statement is true of the empty set), and one that was solved only in the early 1950s.
- Is it hard to prove existence?
- What assumptions do we need?
- As it turns out, economists search for a proof for a long time hit a dead end.
- Until a mathematician came along that inspired them to look at the problem in a different way.

To answer these questions, we study the conditions that characterize an equilibrium.

Existence of A Competitive Equilibrium: Equations and Unknowns

- For some price vector p, suppose we find Walrasian demand $x_i^*(p)$ for each consumer and supply $y_j^*(p)$ for each firm.
- We find an equilibrium if we find a p that solves the following system of L equations:

$$\begin{aligned} \sum_{i=1}^{I} (x_{1i}^*(p) - \omega_{1i}) - \sum_{j=1}^{J} y_{1j}^*(p) &= 0 \\ & \dots \\ \sum_{i=1}^{I} (x_{Li}^*(p) - \omega_{Li}) - \sum_{j=1}^{J} y_{Lj}^*(p) &= 0 \end{aligned}$$

  - In general we may have correspondences, but ignore that for now.
- Since one of the equations is redundant, we can add a normalization (prices sum up to one, or price of one of the goods is equal to one).
- Counting equations and unknowns does not really work, but thinking about the left hand side of this system inspired ideas helpful to find a way to prove that (under some conditions) an equilibrium exists.

Excess Demand

Definition

The market excess demand correspondence $z : \mathbf{R}^L_+ \to \mathbb{R}^L$ is:

$$z(p) = \sum_{i=1}^{I} (x_i^* - \omega_i) - \sum_{j=1}^{J} y_j^*$$

where $y_j^* \in y_j^*(p)$ for each $j = 1, ..., J$, and $x_i^* \in x_i^*(p)$ for each $i = 1, ..., I$.

- For each good, this measures the difference between total demand and total supply.
- If Walrasian demand and firm's supply are single valued, aggregate excess demand is a function.
- Suppose p^* is an equilibrium price vector. Then

$$p^* \cdot z(p^*) = \sum_{l=1}^{L} p_l^* \left[\sum_{i=1}^{I} (x_{li}^* - \omega_{li}) - \sum_{j=1}^{J} y_{lj}^* \right] = 0$$

because at an equilibrium either the terms in square parenthesis are 0, or the corresponding price is equal to 0.

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- If Walrasian demand and firm's supply are single valued, aggregate excess demand is a function.
- Suppose p^* is an equilibrium price vector. Then

$$p^* \cdot z(p^*) = \sum_{l=1}^{L} p_l^* \left[\sum_{i=1}^{I} (x_{li}^* - \omega_{li}) - \sum_{j=1}^{J} y_{lj}^* \right] = 0$$

because at an equilibrium either the terms in square parenthesis are 0, or the corresponding price is equal to 0.

Walras' Law

Theorem (Walras' Law)

In an economy with locally non satisted preference

$$p \cdot z(p) = 0 \qquad \text{for all } p$$

- The value of aggregate excess demand is zero for any price vector.

Proof.

For a fixed price vector p, take any $y_j^* \in y_j^*(p)$ and $x_i^* \in x_i^*(p)$

$$p \cdot z(p) = p \cdot \sum_{i=1}^{I} [x_i^* - \omega_i] - p \cdot \sum_{j=1}^{J} y_j^* = \sum_{i=1}^{I} [p \cdot x_i^* - p \cdot \omega_i] - \sum_{j=1}^{J} p \cdot y_j^*$$

$$\underbrace{=}_{\text{by l.n.s.}} \sum_{i=1}^{I} \left[p \cdot \omega_i + \sum_{j=1}^{J} \theta_{ij} p \cdot y_j^* - p \cdot \omega_i \right] - \sum_{j=1}^{J} p \cdot y_j^*$$

$$= \sum_{j=1}^{J} \sum_{i=1}^{I} \theta_{ij} p \cdot y_j^* - \sum_{j=1}^{J} p \cdot y_j^* \underbrace{=}_{\sum_{i=1}^{I} \theta_{ij} = 1} 0$$

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If $L-1$ markets clear, the L-th market also clears

- If the equilibrium conditions are satisfied in all but one markets, they are also satisfied in the last market.
- The equilibrium conditions can be written as
  $p_l^* z_l(p^*) = p_l^* \left[\sum_{i=1}^{I}(x_{li}^* - \omega_{li}) - \sum_{j=1}^{J} y_{lj}^*\right] = 0$ for each l.
- Suppose $p_l z_l(p) = 0$ for all goods but k. By Walras' Law:
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Excess Demand and Competitive Equilibirum

- A competitive equilibrium says: everyone maximizes (1. and 2.) and

$$\sum_{i=1}^{I}(x_i^* - \omega_i) - \sum_{j=1}^{J} y_j^* \leq 0 \qquad \text{and } p_l^* = 0 \text{ if the inequality for good } l \text{ is strict}$$

- One can re-state this condition using market excess demand:

$$z(p^*) \leq 0 \qquad \text{and } p_l^* = 0 \text{ if the inequality for good } l \text{ is strict}$$

- Hence, we have an equilibrium price vector if we find a price vector such that market excess demand is weakly negative, and prices are zero whenever market excess demand is strictly negative.

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Monotonicity and Zero Prices

- If preferences are strongly monotone (this is more restrictive than local non satiation), then an equilibrium price vector must be strictly positive: if not, consumers would demand an infinite amount of any free good.

With strong monotonicity, a price vector is an equilibrium if and only if it 'clears all markets'

- Formally, a price vector p^* is an equilibrium if and only

$$z_l(p^*) = 0 \quad \text{for all } l,$$

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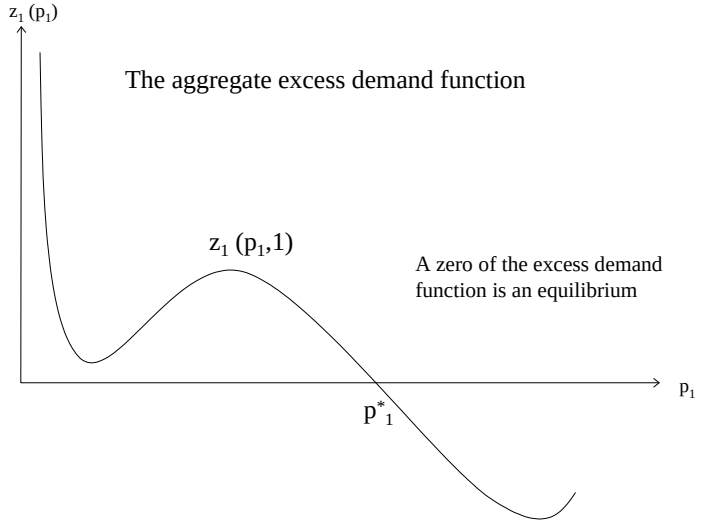
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Equilibrium in an Edgeworth Box

- Assume strictly convex preferences: all excess demands are functions.
- Let the $p_2 = 1$, so we need only find p_1, and write market excess demand as
$$z(p_1, 1) = (z_1(p_1, 1), z_2(p_1, 1))$$
- By Walras' Law, an equilibrium satisfies $z_1(p_1, 1) = 0$ (the other market also clears).
- An equilibrium exists if for some price of good one $z_1(p_1, 1) = 0$.

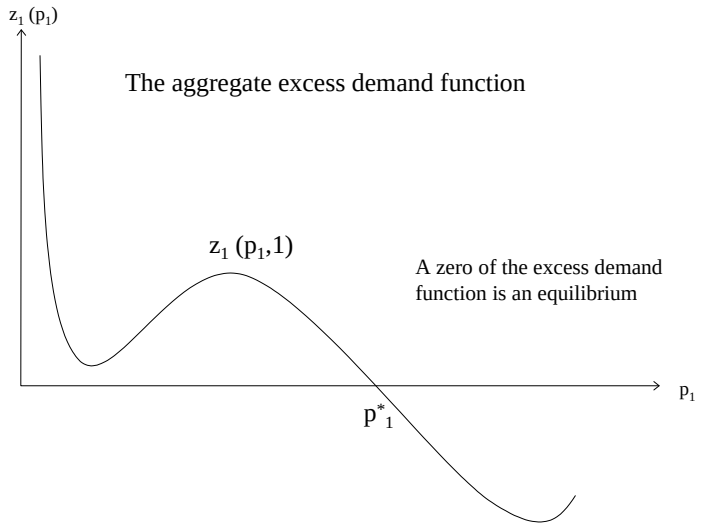


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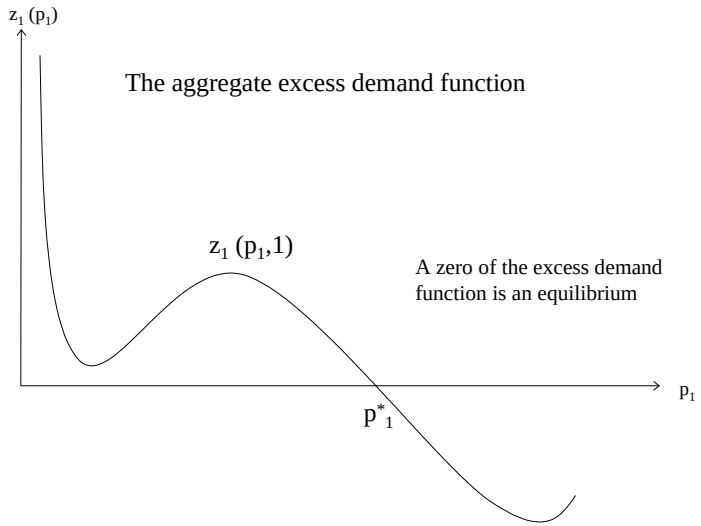
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- With more than 2 goods we need a more general version of this idea.

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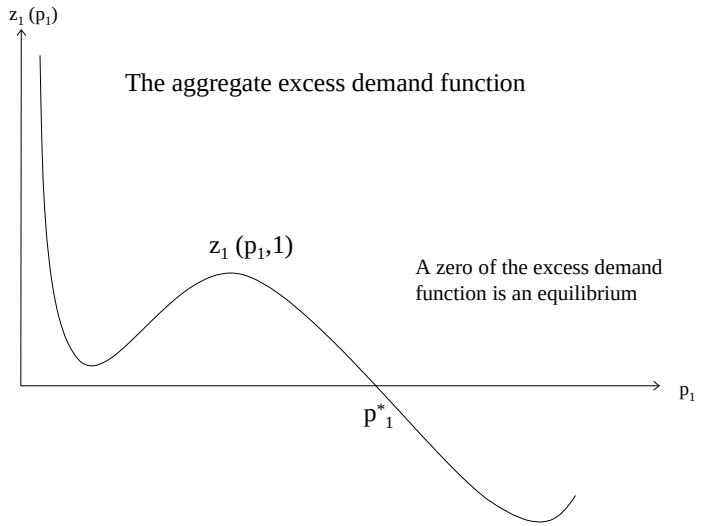
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Next Class

- Existence.